Continuous-Time Radar-Inertial Odometry with B-Spline Sliding-Window Optimization

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***Abstract*—We present a radar-inertial odometry (RIO) system for 6-DoF state estimation on a quadrotor using a single-chip 60 GHz mmWave radar (10–60 points per frame) and a 1 kHz IMU, evaluated through aggressive racing and attitude-tumbling flight up to $10\,\mathrm{rad\,s^{-1}}$: a regime existing RIO evaluations do not cover, where the radar model itself degrades. The trajectory is a continuous-time quintic position B-spline and a cumulative SO(3) orientation B-spline, optimized by a fixed-lag smoother with Schur complement marginalization at a 0.3 s stride over a 3 s window; a sensor-only pipeline initializes it without external *pose* reference. Doppler measures velocity directly, so we evaluate RIO as a *velocity and orientation* source, reporting position as drift per distance. A Markov-blanket factor-selection rule makes the marginalization a true marginal, and a single dynamics-adaptive measurement-weighting law covers hover to $10\,\mathrm{rad\,s^{-1}}$, validated on held-out flights. At the live leading edge (the newest window estimate, before later windows refine it), velocity and orientation RMSE stay below $0.5\,\mathrm{m\,s^{-1}}$ and 3° across racing (position drift under $1\,\%$) and degrade gracefully to 6.4° orientation ($2.9\,\%$ drift) under tumbling; the output orientation covariance is approximately calibrated (NEES) on the racing flights. We also document negative results: a diagnostic for marginalization inconsistency, and that the tumbling elevation bias is a flight-regime proxy, not an antenna constant.**

***Index Terms*—radar-inertial odometry, continuous-time estimation, B-spline, sliding window, marginalization, mmWave radar, Doppler velocity**

I. Introduction

Visual odometry and light detection and ranging (LiDAR)-based simultaneous localization and mapping (SLAM) are well-established, but both degrade in dusty, foggy, featureless, or textureless environments. Millimeter-wave (mmWave) frequency-modulated continuous-wave (FMCW) radar is inherently robust to these conditions: it penetrates smoke and dust, operates in the dark, and measures Doppler velocity directly without feature matching. These properties make it attractive for aggressive flight in unstructured environments.

Combining a mmWave radar with an inertial measurement unit (IMU) (radar-inertial odometry (RIO)) for ego-motion estimation has been explored with discrete-time extended Kalman filter (EKF) and graph-optimization formulations [1], [2]. However, the asynchronous nature of radar-IMU data (radar at $\sim$11 Hz, IMU at $\sim$1 kHz) motivates a continuous-time representation where the trajectory is a smooth function evaluated at any sensor timestamp without interpolation approximation.

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Code and datasets are available at https://github.com/Dr1zz3l/radar-iwr6843-driver.

Research question. This work asks, end to end: *can a single-chip mmWave radar, fused with an IMU, provide usable 6-DoF state estimation, and if so, what does it take?* The starting point is what the sensor can and cannot do: per-point Doppler yields direct ego-velocity, but at 10–60 points per frame there are no bearing-stable features for position fixing, and yaw is a gauge freedom of Doppler+IMU. The system is therefore a *velocity and orientation* source whose position estimate drifts with distance, by construction rather than by defect. What it takes to turn that capability into accurate estimates, and where it ultimately fails, is the subject of the rest of this paper.

We contribute:

- *Single-chip radar under aggressive and tumbling flight*, characterizing how the radar measurement model breaks in this regime: rate-dependent intra-frame smearing, and the finding that the apparent elevation bias under tumbling is a flight-regime error proxy, not a physical antenna constant. Backflips at $10\,\mathrm{rad\,s^{-1}}$ on a single-chip sensor are, to our knowledge, unexplored.
- A *consistency-analyzed* fixed-lag smoother: a factor-set selection rule exploiting B-spline local support to make the Schur marginal exactly closed, distinct from and complementary to first-estimates Jacobian (FEJ), eliminating prior-weight tuning, with a real-time operating point per regime at near-identical accuracy (0.3–0.5 s stride on a laptop).
- One measurement-weighting configuration from hover to $10\,\mathrm{rad\,s^{-1}}$, with the velocity–orientation trade as a measured Pareto curve and accuracy validated on *held-out flights* never used for tuning.
- Honest negative results: a diagnostic symptom pattern for marginalization inconsistency; a representation-level study that *disproves* the adaptive-knot-density hypothesis for this regime; ground-truth limits during aggressive maneuvers; and a normalized estimation error squared (NEES) check confirming the reported orientation covariance is consistent on racing flights.

Data and code. The complete system (physically correct Doppler model with lever arm, sensor-only initialization, full hyperparameter disclosure, and deterministic re-runs) is released for independent reproduction. The datasets (Vicon motion capture (MoCap) lab, TI IWR6843AOPEVM, Pixhawk) and the C++ Ceres solver with Python bindings are available at https://github.com/Dr1zz3l/radar-iwr6843-driver, which despite its driver-style name hosts the complete estimator, solver, and datasets.

II. Related Work

A recent survey [3] catalogues the rapidly growing use of mmWave radar in robotics across motion estimation, localization, and mapping; we focus on the radar-inertial odometry thread most relevant to single-chip aerial fusion.

Radar ego-velocity estimation. Doppler-based ego-velocity estimation from mmWave radar was demonstrated by Kellner et al. [4] and extended to 3D with weighted least squares by Kramer et al. [2], whose RVE system fuses radar with IMU in a discrete-time EKF. Doer and Trommer [1] present an EKF-based RIO with online extrinsic calibration. Our work uses the same per-frame weighted least squares (WLS) ego-velocity estimator but embeds it in a continuous-time batch/sliding-window optimizer.

Continuous-time trajectory estimation. Furgale et al. [5] introduced uniform B-splines for continuous-time visual-inertial calibration. The basalt visual-inertial odometry (VIO) [6] uses cumulative B-splines on Lie groups for stereo-inertial odometry, as does the continuous-time stereo-inertial odometry of Hug et al. [7]. We adapt the basalt-style cumulative SO(3) B-spline to radar-inertial fusion.

Marginalization in sliding-window estimators. OKVIS [8] and VINS-Mono [9] pioneered Schur complement marginalization for visual-inertial systems, propagating information from dropped keyframes as a dense Gaussian prior. We apply the same principle to continuous-time B-spline control points and document the practical challenges of prior scaling when the marginal's orientation and position information differ by ten orders of magnitude.

Closest related work. Continuous-time radar-inertial odometry was introduced by Ng et al. [10], who fuse Doppler measurements from *multiple automotive* radars with an IMU as a least-squares problem over a B-spline trajectory, evaluated on passenger vehicles. Burnett et al. [11] use a Gaussian-process motion prior for spinning-radar- and lidar-inertial odometry, and Štironja et al. [12] model the IMU trajectory continuously inside an RIO estimator with online spatio-temporal calibration. Single-chip radar-inertial estimation has itself received recent attention: Huang et al. [13] exploit radar cross-section and Doppler physics to make a sparse single-chip cloud usable in a discrete-time sliding-window optimization, evaluated on ground-robot and handheld platforms. Our setting differs from all of these in sensor and regime: a *single-chip* radar (10–60 points per frame, 2-TX elevation array) on a quadrotor in aggressive racing and attitude-tumbling flight up to $10\,\mathrm{rad\,s^{-1}}$, a regime in which, as Section VII-C shows, the radar measurement model itself degrades in ways invisible on ground vehicles and in benign aerial navigation.

Single-chip RIO on aerial platforms. Single-chip radar-inertial fusion has been shown on multirotors, but only for benign flight. Michalczyk et al. [14] use a discrete-time EKF with scan-to-scan point matching and Doppler (a factor-graph reformulation follows in [15]); Girod et al. [16] run a real-time onboard M-estimator smoother on a quadrotor in cities and forests, but cap velocity at $2\,\mathrm{m\,s^{-1}}$ and anchor the vertical channel with a barometer. None uses a continuous-time spline, addresses aggressive or tumbling flight, or, as we do, recovers the vertical channel from radar and IMU alone (Section VI-F).

Sliding-window marginalization and consistency. That naïve sliding-window marginalization is inconsistent is long established in discrete time [17], where FEJ methods fix linearization points to preserve the observability nullspace. In continuous time, CLIC [18] and LIO-MARS [19] marginalize spline knots leaving the window by adjacency to the dropped knots; this is correct when the estimated states are all local to the window, but not when a *global* shared IMU bias couples every IMU factor and tempts the adjacency rule to pull in far-edge factors, producing an over-confident prior. Our contribution, narrower than and complementary to FEJ, is a factor-*set* selection rule that exploits B-spline local support to keep the marginal exactly closed despite this shared bias, together with a diagnostic symptom pattern for the inconsistent variant; both are developed in Section IV-E and Section VII-D. In the discrete-time UAV line, the EKF-RIO family of Doer and Trommer [1] defines the state of practice. Absolute root-mean-square error (RMSE) is not comparable across papers: our own quantization ablation (Section VI-E) shows firmware configuration alone changes Doppler resolution by $12\times$, so rather than quote their numbers we cross-validate against this line directly: Section VI-F runs our system on the public ICINS-2021 flight datasets of [20] and re-evaluates their EKF estimators under our causal metrics on the same data, and reports the regime-dependent result of porting their filters to our platform.

III. System Overview

A. Problem Formulation

Estimation problem. We estimate the continuous-time 6-DoF body trajectory (world-frame position $\mathbf{p}_w(t) \in \mathbb{R}^3$ and orientation $\mathbf{R}(t) \in \mathrm{SO}(3)$) together with the constant IMU biases $\mathbf{b}_a, \mathbf{b}_g$, from asynchronous measurements over the estimation interval (the full trajectory in batch mode, a sliding window online): radar Doppler velocities ($\sim 11\,\mathrm{Hz}$), accelerometer and gyroscope samples ($1\,\mathrm{kHz}$), and a low-rate heading prior. We adopt a *continuous-time* representation because the sensors are asynchronous and run at disparate rates: a trajectory defined for every t is evaluated at each measurement's exact timestamp, and its derivatives (velocity $\dot{\mathbf{p}}_w$, acceleration $\ddot{\mathbf{p}}_w$, and body angular velocity $\boldsymbol{\omega}_b$) follow in closed form, so no inter-sample interpolation or rigid-keyframe approximation is required.

Parameterization. The trajectory is a B-spline (Section IV-A): the position is a quintic uniform B-spline with control points $\{\mathbf{P}_i\} \subset \mathbb{R}^3$, and the orientation a cumulative $\mathrm{SO}(3)$ B-spline with incremental control points $\{\boldsymbol{\Omega}_j\} \subset \mathfrak{so}(3)$. This renders the otherwise infinite-dimensional trajectory finite-dimensional; the control points and the IMU biases form the estimated state

$$\mathcal{X} = \{\, \mathbf{P}_i,\ \boldsymbol{\Omega}_j,\ \mathbf{b}_a,\ \mathbf{b}_g \,\}. \tag{1}$$

The batch solve additionally frees the single observable extrinsic degree of freedom (the radar pitch); the sliding window fixes it at the measured value (Section IV-A).

Estimator. Treating the measurement noise as zero-mean and independent, the maximum-a-posteriori (MAP) estimate is the minimizer of the negative log-posterior, a weighted nonlinear least-squares problem over the measurement residuals, the spline regularizers, and the priors:

$$\hat{\mathcal{X}} = \arg\min_{\mathcal{X}} \sum_k \rho_\delta\big(r_{D,k}^2\big) + \lambda_a \sum_m \|r_{a,m}\|^2 + \lambda_g \sum_m \|r_{g,m}\|^2 + \lambda_\psi \sum_n r_{\psi,n}^2 + E_{\text{reg}}(\mathcal{X}) + E_{\text{prior}}(\mathcal{X}), \tag{2}$$

where r_D, r_a, r_g, r_ψ are the radar-Doppler, accelerometer, gyroscope, and heading residuals of Section IV-B; ρ_δ is the Huber kernel (δ=1.0 m/s) that bounds the influence of heavy-tailed Doppler outliers; the $\lambda_\bullet$ are the (rate-adaptive) information weights of Section IV-B; E_{reg} collects the position min-snap and orientation smoothness regularizers (Section IV-C); and E_{prior} holds the boundary and, online, the marginalization priors (Section IV-E). We minimize (2) with Levenberg–Marquardt (Ceres) using analytic Jacobians. The fixed-lag estimator solves it over a 3 s window and, on each stride, marginalizes the states leaving the window into E_{prior} via the Schur complement (Section IV-E), keeping the per-window cost bounded.

B. Hardware Platform

The system runs on a quadrotor with a TI IWR6843AOPEVM mmWave radar (60–64 GHz FMCW), a Pixhawk flight controller IMU, and a Vicon MoCap system used for ground truth evaluation. The radar is mounted upside-down at a nominal 30° downtilt by computer-aided design (CAD); the as-built angle is a few degrees lower (Section VI), and the extrinsic rotation is $[180°, 27.5°, 0°]$ (roll, pitch, yaw), translation $[0.08, 0.02, -0.01]$ m in the body frame. The radar boresight lies along the body forward axis, so the Euler pitch is numerically the physical boresight downtilt. The radar operates at $\sim$11 Hz with 0.049 m/s Doppler resolution (best-velocity firmware configuration). Raw IMU rate is $\sim$993 Hz; both modalities are used at full rate.

C. Pipeline

Fig. 1 shows the pipeline. A three-stage sensor-only initialization (P1–P3, detailed in Section IV-D) builds the initial trajectory the nonlinear solve needs, from radar and IMU data alone; the same raw measurements then also enter the solver directly as factors (Fig. 1, dashed). MoCap supplies only the yaw heading prior (a magnetometer surrogate, Section IV-B) and the final RMSE evaluation; P1–P3 and the position/attitude estimate use no external pose reference.

IV. Methodology

A. State Representation

Position. The world-frame position $\mathbf{p}_w(t) \in \mathbb{R}^3$ is a quintic (order-6) uniform B-spline with knot spacing $\Delta t_p = 5$ ms (slow racing; per-regime grids in Table II). Velocity $\dot{\mathbf{p}}_w$ and acceleration $\ddot{\mathbf{p}}_w$ are obtained as analytical derivatives of the spline.

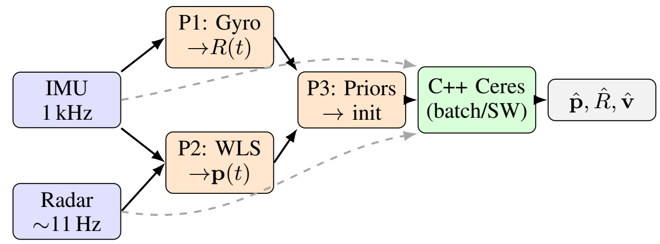

Fig. 1. System pipeline. Solid: initialization flow (P1–P3 seed the optimizer). Dashed: raw sensor data fed as measurement factors. Radar returns first pass a RANSAC ego-velocity prefilter (Section IV-B) that rejects elevation-biased outliers before both P2 and the solver. MoCap supplies only the yaw heading prior (magnetometer surrogate) and RMSE evaluation.

Orientation. We use the *cumulative* SO(3) B-spline formulation [21]:

$$\mathbf{R}(t) = \mathbf{R}_{\text{base}}[k{-}3] \prod_{j=k-3}^{k} \operatorname{Exp}\big(\tilde{B}_j(t)\,\boldsymbol{\Omega}_j\big), \tag{3}$$

where k is the active knot index for time t, $\boldsymbol{\Omega}_j \in \mathfrak{so}(3)$ are incremental rotation control points (standard Exp/Log map convention), $\tilde{B}_j(t)$ are cumulative cubic basis functions, and $\mathbf{R}_{\text{base}}$ is the left anchor rotation. Knot spacing is $\Delta t_o = 8$ ms (slow racing; see Table II). Body-frame angular velocity $\boldsymbol{\omega}_b(t)$ is obtained as a closed-form derivative of (3) directly from `evaluate_lie()` without numerical differentiation.

Biases and extrinsics. A constant 6-DoF IMU bias $[\mathbf{b}_a, \mathbf{b}_g]$ is estimated jointly. The extrinsic pitch δ_{pitch} (1 DoF) is optimized with a soft prior; roll and yaw are fixed (unobservable from Doppler). It is freed in the batch solve but locked at the measured 27.5° for the sliding window (Section VII-C).

B. Measurement Models

Radar Doppler with lever arm. The TI IWR6843 reports positive Doppler for a receding target. With antenna offset $\mathbf{t}_{bs}$ from the body center of mass (CoM), the predicted Doppler for a point at sensor-frame bearing $\hat{\mathbf{u}}_s$ is:

$$v_D = -\hat{\mathbf{u}}_b^\top\big(\mathbf{R}_{bw}\,\mathbf{v}_w(t) + \boldsymbol{\omega}_b(t) \times \mathbf{t}_{bs}\big)\,, \tag{4}$$

where $\hat{\mathbf{u}}_b = \mathbf{R}_{bs}\,\hat{\mathbf{u}}_s$, $\mathbf{R}_{bs}$ is the rotation from radar sensor to body frame (extrinsic [roll=180°, pitch=27.5°, yaw=0°]; measured, Section VI), and $\mathbf{R}_{bw} = \mathbf{R}_{wb}^\top$. The leading minus makes the convention self-consistent: a sensor translating toward a static point ($\hat{\mathbf{u}}_b^\top \mathbf{R}_{bw}\mathbf{v}_w > 0$) yields $v_D < 0$, an approaching rather than receding target, as required. The residual $r_k = v_{D,\text{meas}} - v_{D,\text{pred}}$ uses Huber loss with $\delta = 1.0$ m/s, robust as defense-in-depth to any residual Doppler noise on the surviving inlier returns; the $0.049\,\mathrm{m\,s^{-1}}$ Doppler resolution sets a lower bound but does not drive the choice.

RANSAC ego-velocity prefilter (default front-end). A single-chip radar with only 2 TX antennas has limited elevation diversity, and a minority of returns carry an elevation-correlated Doppler bias that a robust kernel only down-weights. Following reve [1] (Doer's radar ego-velocity estimator), we therefore prepend a reve-style 3D least-squares RANSAC ego-velocity estimate to the per-frame Doppler set (inlier threshold $0.15\,\mathrm{m/s}$) and pass only the inlier returns to the spline solver, hard-rejecting the elevation-biased minority before the optimization. The prefilter is seeded (`numpy.default_rng(0)`), so the full pipeline reproduces bit-identically; frames with fewer than five returns bypass it unchanged. On those sparse frames, including most of the backflip regime, the in-solver Huber loss is the sole radar outlier protection, so the two stages are complementary, not redundant. It runs once at frame load, not per window, so it does not enter the per-window timing budget (Section VI). Section VI-F quantifies what it removes: on the public ICINS flights it closes an order-of-magnitude vertical-drift gap, and on our own flights it improves aggressive-racing live position by 20% ($0.50 \rightarrow 0.40\,\mathrm{m}$ live). It is enabled by default for every result in this paper.

Dynamics-adaptive measurement weighting. The central practical finding of this work is that a single *rate-adaptive trust allocation* across the three measurement streams covers hover to $10\,\mathrm{rad\,s^{-1}}$ tumbling with one configuration. Each component was derived from a measured failure diagnosis (Section VII-C), not tuned ad hoc:

- *Gyroscope: trust more when rotating fast.* Per-sample weight $\lambda_g^{\text{eff}} = \lambda_g\big(1+(|\mathbf{z}_g|/\omega_g)^p\big)$ with λ_g=4, ω_g=4 rad/s, p=4, using the *measured* rate (causal, no spline evaluation). The quartic exponent keeps racing untouched ($\lambda^{\text{eff}} \approx \lambda_g$ below 3 rad/s) and reaches $\lambda^{\text{eff}} \approx 160$–500 during flips, where the gyroscope is the only clean sensor. A quadratic ramp (p=2) stiffens the gyro already at 2–5 rad/s (where radar is still informative) and degrades racing roll/yaw.
- *Radar: trust less when rotating fast.* A radar frame integrates over tens of milliseconds; at body rate ω the scene smears by $\omega\,T_{\text{frame}}$ (≈17° at 10 rad/s). Per-frame noise inflation $\sigma^2_{\text{eff}} = \sigma_0^2\big(1 + (|\boldsymbol{\omega}|/\omega_r)^2\big)$, ω_r=4 rad/s, $|\boldsymbol{\omega}|$ from the warm-start spline at problem-build time; continuous, no hard-gate tuning cliff.
- *Accelerometer: same form,* ω_a=8 rad/s: during flips the accelerometer carries thrust transients and vibration, its gravity information is nil mid-flip, and it otherwise distorts orientation through the $\mathbf{R}$–$\ddot{\mathbf{p}}$ coupling.
- *Per-point signal-to-noise ratio (SNR) weighting.* $w_i = \mathrm{clip}\big((I_i/\tilde{I})^{\alpha},\,0.25,\,4\big)$ with frame-median intensity $\tilde{I}$ and α=1 (median-relative, so the global radar weight is unchanged). High-SNR returns carry measurably better Doppler: on aggressive racing this alone improves live orientation from 3.31° to 2.88° at unchanged position.

The weighting trades are made explicit in Section VII-C: on tumbling flight the radar/accelerometer gates exchange velocity accuracy for orientation accuracy along a measured Pareto curve; (ω_r, ω_a)=$(4, 8)$ is its knee.

Flip-regime Doppler correction. Under sustained tumbling an additional attitude-correlated Doppler systematic appears, which we absorb with a per-point term $v_{\text{corr}} = v_{\text{meas}} - b\,u_{s,z}$ (b=$-1.5\,\mathrm{m/s}$, backflips only). Section VII-C shows why b must be interpreted as a flight-regime error proxy rather than a physical elevation bias of the 2-TX array.

Accelerometer.

$$\mathbf{r}_a = \mathbf{z}_a - \mathbf{R}_{bw}(\ddot{\mathbf{p}}_w - \mathbf{g}_w) - \mathbf{b}_a, \tag{5}$$

weighted $\lambda_a = 0.01$.

Gyroscope.

$$\mathbf{r}_g = \mathbf{z}_g - \boldsymbol{\omega}_b(t) - \mathbf{b}_g, \tag{6}$$

weighted $\lambda_g = 4.0$ (L2 loss).

Gravity direction.

$$\mathbf{r}_{\text{tilt}} = \frac{\mathbf{z}_a - \mathbf{b}_a}{\|\mathbf{z}_a - \mathbf{b}_a\|} - \mathbf{R}_{bw}^{\top}\hat{\mathbf{g}}, \tag{7}$$

active when $\big|\,\|\mathbf{z}_a - \mathbf{b}_a\| - g\,\big| < 3\,\mathrm{m/s^2}$. This provides roll/pitch anchoring during near-hover phases without contaminating dynamic flight.

Heading prior. Yaw is unobservable from Doppler alone (all yaw-equivalent trajectories predict identical radial velocities under pure rotation about gravity). We supply a heading reference from a magnetometer (standard on most platforms; here simulated by MoCap yaw). The residual $r_\psi = \psi_{\text{est}} - \psi_{\text{ref}}$ is added at ∼100 Hz, with $\lambda_\psi = 0.6$ by default and raised to 10 on the heading-critical bags (slow racing, backflips; Table III footnote). A weight sweep shows the split is a yaw-observability/position trade. On slow racing, gentle dynamics leave yaw radar-unconstrained, so the weak default (λ_ψ=0.6) lets yaw drift to 4.3° (orientation 2.0 → 4.2°) while a strong prior costs almost no position. On fast racing, the rich translational excitation already holds yaw near 3.9° at λ_ψ=0.6, so raising it to 10 buys only 0.6° of yaw yet degrades settled position (0.31 → 0.43 m) through the orientation–velocity coupling; we therefore keep it light. On backflips the anchor instead stabilizes the whole attitude through the flip: dropping λ_ψ to 0.6 inflates settled orientation from 5.0° to 6.8° (all three axes degrade), so it stays at 10. The per-regime weight is therefore not monotone in dynamics, which is why it cannot be replaced by an $|\boldsymbol{\omega}|$-keyed rule. λ_ψ is thus the one weight *not* folded into the runtime-adaptive law above, and at present it is set once per flight type rather than auto-weighted. It cannot key off $|\boldsymbol{\omega}|$, since gentle (slow racing) and tumbling (backflips) flight need the same strong prior at opposite rates while translation-rich fast racing needs a weak one; keying it instead off a yaw-observability proxy such as translational excitation, and adapting it to real magnetometer noise in place of the idealized MoCap heading, is left to future work.

C. Regularization

Position: minimum snap. $E_{\text{snap}} = \lambda_s \int \|\mathbf{p}^{(4)}(t)\|^2\,dt$ with $\lambda_s = 2{\times}10^{-5}$.

Orientation: minimum angular acceleration. Rather than penalizing angular velocity increments (minimum ω), which opposes every banked turn and the entire backflip maneuver, we penalize the *second* finite difference on SO(3):

$$\mathbf{r}_\alpha = \mathrm{Log}(\mathbf{R}_{i-1}^{-1}\mathbf{R}_i) - \mathrm{Log}(\mathbf{R}_i^{-1}\mathbf{R}_{i+1}), \tag{8}$$

which is zero for constant angular rate. Swept $\lambda_\alpha \in \{0, 0.001, 0.01, 0.1, 1.0\}$; 0.1 was best across all bags (1.0 caused backflip divergence).

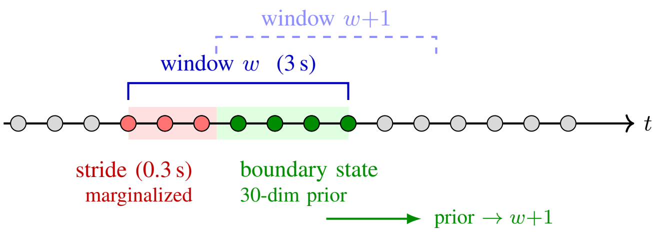

Fig. 2. Sliding window: red CPs (stride zone) are marginalized into a 30-dim Gaussian prior on the green boundary CPs.

D. Sensor-Only Initialization (P1–P3)

P1: Orientation. A stationary segment before flight is detected automatically; the gyroscope bias is the mean reading over this segment and the accelerometer bias a gravity-aligned scalar correction, both *sensor-only* and using no external attitude reference. Initial roll/pitch are then derived from the stationary accelerometer (gravity direction). Yaw is set to zero (gauge freedom; corrected later by the heading prior). The full orientation trajectory is obtained by forward-integrating debiased gyroscope readings: $\mathbf{R}(t+\Delta t) = \mathbf{R}(t)\,\mathrm{Exp}\big((\mathbf{z}_g - \mathbf{b}_g)\Delta t\big)$.

P2: Position. Per radar frame, weighted least squares estimates the 3D ego-velocity $\mathbf{v}_{\text{wls}}$ from the RANSAC inlier returns (Section IV-B). Doppler alias unwrapping uses IMU-integrated world-frame velocity to select the correct alias once at frame load (20.9% of points aliased in fast racing, 1.7% in slow racing); the unwrapped Doppler is the measurement consumed by *both* this WLS estimate and the main solver's radar factors (Eq. 4), with the solver's own $v_{\max}$ check kept only as a residual safety net. World-frame velocity is integrated with forward Euler to produce an initial position trajectory.

P3: Boundary priors. Position, velocity, orientation, and angular velocity boundary priors are built from the sensor-only state at $t=0$. $\lambda_{\text{bnd}} = 1000$ for position and velocity.

E. Sliding Window with Schur Complement Marginalization

The fixed-lag smoother advances a 3 s window in 0.3 s strides. After each Ceres LM solve, the *stride zone* CPs and orientation knots are marginalized:

$$S = H_{bb} - H_{ab}^{\top} H_{aa}^{-1} H_{ab}, \tag{9}$$

where a indexes stride-zone variables and b indexes boundary variables (last $N-1$ position CPs, last $N-1$ orientation knots, and the 6-DoF bias; dimension 30). S is factored as $S = LL^{\top}$ and stored as a `MargPriorFunctor` for the next window.

Re-centered prior. Before adding the prior to each window's problem, the linearization point $\mathbf{x}_0$ is updated to the current warm-start. This makes the prior contribute only curvature (the Hessian shape), not a gradient pull toward a stale historical estimate. The residual is $\mathbf{r} = L^{\top}(\mathbf{x} - \mathbf{x}_0)$ in local coordinates.

Consistent factor selection (Markov-blanket rule). Which residuals enter Eq. (9) determines whether the prior is a true marginal. A naïve choice (all residuals touching marginalized *or* boundary variables) pulls in every IMU factor of the window through the shared bias block ($\sim$25,000 residual rows), *conditions* on interior states (treating them as perfectly known), and double-counts factors that are re-added in the next window. The resulting prior is overconfident by orders of magnitude and must be deflated by a hand-tuned scale of 10^{-7}–2×10^{-4} depending on flight dynamics (Section VII-D). The shared global bias is the trap: it is what tempts pulling in the far-edge IMU factors. We instead exploit the splines' local support of N knots (quintic position N=6, cubic orientation N=4).

Proposition 1 (Exactly closed spline marginal)**.** *Let $\mathcal{M}$ be the block of knots leaving the window and $\mathcal{B}$ the $N-1$ retained knots immediately adjacent to it. For an order-N spline, every factor has local support over N consecutive knots $[i,\, i+N-1]$. If a factor touches $\mathcal{M}$, i.e. $[i,\, i+N-1] \cap \mathcal{M} \neq \varnothing$, then $[i,\, i+N-1] \subseteq \mathcal{M} \cup \mathcal{B}$. Selecting exactly the factors that touch $\mathcal{M}$, together with the previous prior, therefore forms a Schur complement that conditions on no interior state and shares no factor with the next window: the resulting prior is the exact marginal.*

Including *only* residuals that touch a marginalized variable (rather than those touching marginalized *or* boundary variables) thus yields a consistent marginal. With this rule the prior is applied **unscaled** (scale $=1$) across all flight regimes, and its Mahalanobis residual at the solution drops from 10^{6}–10^{12} to $\mathcal{O}(1)$: the prior agrees with the incoming data.

Relation to FEJ. This mechanism is distinct from, and complementary to, first-estimates-Jacobian approaches to marginalization consistency [17]: FEJ fixes the *linearization points* of the prior to preserve the observability nullspace, whereas our rule corrects the *factor set* entering the Schur complement so that the marginal is exactly closed *at a fixed linearization point*. We do not employ FEJ; instead the prior is re-centered at each window's warm start (curvature retained, gradient pull discarded). We do not claim this re-centering preserves consistency across relinearizations as a theoretical guarantee; rather, in our experiments it left no measurable inconsistency symptom; the NEES analysis of Section VI-G provides the corresponding empirical evidence. Our missions are short (18–26 s), which may be too brief to expose drift a re-centered prior could accumulate over a long deployment; bounding this over extended runs is left to future work.

Live-edge warm start. Control points entering the window each stride would otherwise hold stale dead-reckoned initialization values; any CP-level seam is amplified by the min-snap term ($\propto \Delta/\Delta t_p^4$, initial cost $\sim10^{13}$). We rigidly align the entering segment (shift + ΔR) to the last data-supported CP of the previous solution, reducing the initial cost by five orders of magnitude.

Marginalization cost. The restricted Gauss–Newton Hessian for Eq. (9) is accumulated by evaluating the affected cost functions directly (tangent-space Jacobians via the manifold plus-Jacobian, Triggs correction for robust losses), avoiding a full solver re-linearization; the marginalization step costs $\sim$10 ms per window.

TABLE I
DATASET SUMMARY

Bag	Dur. (s)	$v_{\max}$ (m/s)	$\|\omega\|_{\max}$ (rad/s)	Frames
Slow racing	25.8	3.65	2.62	258
Fast racing	18.0	5.66	9.14	179
Backflips*	18.1	∼4.3 (95%)	∼10.9 (95%)	181

*95th percentile; MoCap tracking artifacts dominate the instantaneous peaks. The ∼10 rad/s headline is this 95th-percentile sustained rate, not an instantaneous maximum.

TABLE II
PER-REGIME SLIDING-WINDOW CONFIGURATION

Parameter	Slow	Fast	Backflips
Position knot Δt_p (ms)	5	40	10
Orientation knot Δt_o (ms)	8	16	8
LM iteration cap	12	400	400
Heading prior λ_ψ	10	0.6	10
Doppler elevation corr. b (m/s)	—	—	-1.5
Position-init anchor λ_{p_0}	—	—	0.5

Shared across all regimes: 3 s window, 0.3 s stride, unscaled Markov-blanket marginalization prior, the single dynamics-adaptive weighting law (Section IV-B), RANSAC ego-velocity prefilter ($0.15\,\mathrm{m\,s^{-1}}$ inlier), sensor-only initialization, and frozen extrinsic pitch $27.5°$. "—" marks the shared default: uncapped-by-data iterations, and the flight-regime elevation correction / position tether off.

V. EXPERIMENTAL SETUP

All experiments use the rosbag datasets collected in a Vicon MoCap lab. Table I summarizes the three bags used for evaluation.

All three regimes run the *same* sliding-window solver, weighting law, and front-end; Table II collects the handful of per-regime knobs that differ, so that every reported number can be reproduced from one place.

IMU rate: ∼993 Hz (full rate, no downsampling for C++ solver). **Evaluation:** Start-anchored rigid alignment to MoCap ground truth: the alignment rotation $\mathbf{R}_{\text{align}} = \mathbf{R}_{\text{gt},0}\,\mathbf{R}_{\text{est},0}^{\top}$ and translation $\mathbf{t}_{\text{align}} = \mathbf{p}_{\text{gt},0} - \mathbf{R}_{\text{align}}\,\mathbf{p}_{\text{est},0}$ are fixed at frame 0 (no scale). The final 3 s is trimmed. Velocity RMSE is computed from the analytical first derivative of the position B-spline, not from finite differences of the estimated trajectory. *Position drift* (%) is position RMSE normalised by the total GT path length, the standard odometry metric for a dead-reckoning system with no loop closure. Start-anchored alignment is harsher than the whole-trajectory SE(3)/Umeyama fit (the EVO/KITTI default) used by most odometry papers, because it cannot retro-fit a constant transform to cancel accumulated drift: it fixes the frame once, at the start, exactly as a deployed estimator must. We adopt it precisely for this causal, deployment-relevant reading, and apply it identically to every system compared. Where we compare against externally published figures (Section VI-F) we additionally report whole-trajectory-aligned numbers so the baselines can be cross-checked against their own papers. *Translational relative pose error (RPE)* (relative pose error, Fig. 5) is the KITTI-standard per-segment displacement error averaged over all sub-segments of a given accumulated GT length. Because the start-anchored alignment offsets cancel within every relative displacement, RPE is alignment-independent. For sliding window, *live (leading-edge)* RMSE is computed by snapshotting the estimate at $[t_{\text{end}}-\text{stride},\, t_{\text{end}}]$ of each window at 50 Hz before any subsequent refinement. The *settled* RMSE evaluates the same trajectory after all windows have refined each point.

TABLE III
SLIDING WINDOW: SETTLED AND LIVE LEADING-EDGE RMSE
(CONSISTENT UNSCALED PRIOR; PER-WINDOW SOLVE TIME ON A LAPTOP CPU)

Bag	Mode	Vel (m/s)	Ori (°)	Pos (m)	Drift%	t_{win}(s)
Slow racing	Settled	0.42	1.53	0.293	0.61	0.70
	Live edge	**0.48**	**1.97**	**0.310**	**0.64**	
Fast racing	Settled	0.25	2.35	0.312	0.69	0.35
	Live edge	**0.32**	**2.88**	**0.399**	**0.88**	
Backflips‡	Settled	—*	5.01	1.674	3.09	0.6
	Live edge	**2.35**	**6.35**	**1.545**	**2.85**	

All bags use the unscaled consistent marginalization prior (Section IV-E) and the *same* dynamics-adaptive weighting law (Section IV-B). Per-regime parameters are limited to spline grids ($\Delta t_p{=}5/40/10$ ms), the heading weight, and for backflips the position-init anchor ($\lambda_{\text{pos-init}}{=}0.5$) and flip-regime Doppler correction $b{=}-1.5$ m/s (Section VII-C). Extrinsic pitch locked at 27.5° for all bags. *Settled velocity is not reported for backflips: stride-boundary resampling injects a retrospective-evaluation artifact there. The live column is the deployment-relevant figure throughout. Drift %: path lengths 48.1 m / 44.9 m / 54.1 m. Iteration-capped configurations reproduce bit-identically across repeated runs; full-convergence configurations vary by <2% (multithreaded reduction order).

Solver implementation. The Ceres LM problem is solved with `SPARSE_NORMAL_CHOLESKY` (SuiteSparse), up to 400 iterations. All three factor types (gyroscope, accelerometer, and radar Doppler) use hand-derived analytic Jacobians based on the basalt cumulative-spline Jacobian struct; Ceres automatic differentiation was eliminated to avoid Jet-arithmetic overhead over the large set of spline control-point parameters per factor. In the sliding-window solver the linear solve and the marginalization-prior recompute (`compute_prior`, evaluated *every* LM iteration, as distinct from the ∼10 ms one-time prior *formation* of Section IV-E) are the dominant costs; Jacobian evaluation accounts for roughly 25–30% of per-window wall time.

VI. RESULTS

A. Batch Solver and Extrinsic Pitch

We use the full-trajectory C++ batch solve as an offline full-batch reference; offline it reaches $0.20\,\mathrm{m} / 1.1°$ on slow racing and $0.64\,\mathrm{m} / 2.5°$ on fast (the dense racing Δt_p over-fits fast racing, where the coarser sliding window does better). The radar-to-body pitch is a fixed mount angle, measured at 27–28° with an inclinometer on the radar face (reference surface 0°) and frozen at 27.5°, a few degrees below the 30° CAD design (3D-printed, hand-assembled fixture). We lock it rather than estimate it online because it is only weakly observable from Doppler: the self-calibration is init-dependent and orientation RMSE is pitch-insensitive (Section VII-C). The batch estimate is shown for reference (dashed) in the RPE curves (Fig. 5).

B. Sliding Window: Settled vs. Live

Table III reports sliding window results (3 s window, 0.3 s stride, Schur marginalization). Fig. 3 shows trajectory overlays for all three flights; Fig. 4 shows the corresponding error timeseries.

TABLE IV
EVALUATION CONFIGURATION (3 S/0.3 S, USED THROUGHOUT THE PAPER) VS. THE PER-REGIME REAL-TIME POINT (SOLVE TIME BELOW THE STRIDE). REACHING REAL TIME IS ACCURACY-NEUTRAL AND IMPROVES BACKFLIPS. LIVE-EDGE RMSE; SOLVE TIMES CARRY $\sim$20% RUN-TO-RUN VARIANCE.

Regime	Win./str. (s)	Solve (s)	Live pos (m)	Live ori (°)
Fast	3.0/0.3 (eval.)	0.32	0.399	2.88
	2.0/0.3 (real-time)	0.28	0.443	2.95
Slow	3.0/0.3 (eval.)	0.72	0.310	1.97
	1.5/0.5 (real-time)	0.40	0.301	1.98
Backflips	3.0/0.3 (eval.)	0.59	1.545	6.35
	2.0/0.5 (real-time)	0.43	**1.499**	**5.56**

Live-edge orientation degrades 0.4–0.5° relative to settled; this is expected because subsequent windows refine previous estimates. With the consistent prior, settled velocity is smooth across stride boundaries (the earlier inconsistent prior produced position jumps that inflated the settled velocity derivative). We report all tables, plots, and the cross-validation of Section VI-F at a single *evaluation configuration* (3 s window, 0.3 s stride), chosen for best accuracy and cross-table consistency rather than as a deployment claim: per-window solve time is set by window length and knot density, not the stride, and at 3 s only fast racing approaches the 0.3 s budget on a laptop CPU. Real time is nonetheless within reach. Pushing the window and stride to *barely* reach it per regime costs almost nothing (Table IV): fast is real-time at the 0.3 s stride with a 2 s window, slow and backflips at a 0.5 s stride (1.5 s and 2 s windows), live accuracy stays within run-to-run noise of the evaluation configuration (backflips even improves, $6.35 \rightarrow 5.56°$), and racing remains under 1% drift. Solve times carry $\sim$20% run-to-run variance, so the strides are taken with margin.

Deployment, though, asks for more than *barely* real-time. This estimator is designed as one factor source in a larger fusion graph (Section VI-G), sharing compute with a parallel visual or lidar pipeline, so a tuned, no-margin operating point on a dedicated CPU is a feasibility demonstration, not a deployment budget. The margin must come from the back-end: a banded Schur factorization is numerically unstable at our conditioning, so the route is incremental smoothing, not batch re-solve. Girod et al. [16] reach comfortable onboard rates with exactly such an iSAM2 back-end (on a discrete-time graph); porting our continuous-time B-spline onto it, where the shared global bias induces a Bayes-tree fill-in, is the deployment path we leave to future work. The dominant cost reductions were: the consistent prior removing the marginalization re-linearization ($\sim$0.6 s), coarsening Δt_p from 5 ms to 40 ms for aggressive flight (which *also* drops the LM iteration count from 28 to 11 and slightly improves orientation), and warm-start alignment removing the live-edge seam. For a mapping-oriented profile (full iterations, no warm-start alignment) slow racing settles to 1.14° orientation (yaw 0.61°) at 2.1 s per window.

C. Relative Pose Error and Position Drift

Fig. 5 shows the KITTI-style translational RPE as a function of segment length for racing bags (batch and SW live) and backflips (batch only). At longer segments ($\geq$ 20 m) the global B-spline smoother reduces error sharply: slow racing batch reaches 1.08 % at 20 m and 1.02 % at 40 m; fast racing stabilises at $\approx$2.2 % beyond 20 m. At short segments ($\leq$10 m) errors are larger (3–8% for racing) because the 11 Hz radar provides only a handful of Doppler measurements per segment and the global optimizer cannot smooth over sub-second intervals, a constraint of the radar rate, not a spline artifact. Global position drift (%) remains the headline deployment metric (live edge); RPE is the complementary per-segment view.

D. Radar Contribution

Radar is essential. Disabling it (`--no-radar`) collapses the live-edge estimate to IMU dead-reckoning: position drift balloons to 5.9% on slow racing and diverges entirely on fast racing ($\approx$99.5% of distance, error on the order of the full path length), with velocity RMSE rising to 2.4 and 8.0 m/s as double-integrated accelerometer noise accumulates unbounded. The direct Doppler velocity constraints bound this to the Table III figures (0.6–0.9% drift, 0.32–0.48 m/s velocity), one-to-two orders of magnitude lower. Orientation improves more modestly, from 8.9°/4.0° (IMU-only) to 2.0°/2.9°, the 1 kHz gyroscope already constraining it.

E. Held-Out Validation

All tuning used three bags (slow/fast racing, backflips); the final configuration was then run *unmodified* on flights never used for any decision. A second, independent fast-racing flight reaches 0.37 m / 2.81° live (vs. 0.40 m / 2.88° on the tuning bag), and a circular trajectory reaches 0.90 m / 3.77° live. Five additional stress-tier bags from an earlier recording day use an older radar firmware with 12$\times$ coarser Doppler quantization (0.604 vs. 0.049 m/s), partially unusable accelerometer axes, and per-day time offsets; the default RANSAC prefilter still keeps 46–75% of returns (no starvation), and the system degrades gracefully (1.5–7 m position), with the *old-firmware backflips* orientation reaching 8.1° settled / 9.1° live (better than the 14.8° of its previous, individually tuned best) under the untuned universal configuration. The rate-adaptive weighting thus transfers across both firmware and day. A data-health screening (radar frame-rate continuity, IMU/MoCap coverage) precedes all held-out evaluation, since several early bags contain radar universal asynchronous receiver/transmitter (UART) dropouts; all evaluation windows lie on healthy spans. Each flight condition is a single representative recording; the evaluation's breadth comes from these held-out flights and the cross-platform validation of Section VI-F rather than repeated trials (Section VII-A).

F. External Cross-Validation Against EKF Baselines

We cross-validate against the EKF-RIO family of Doer and Trommer [1], [20] in both directions on shared data, with

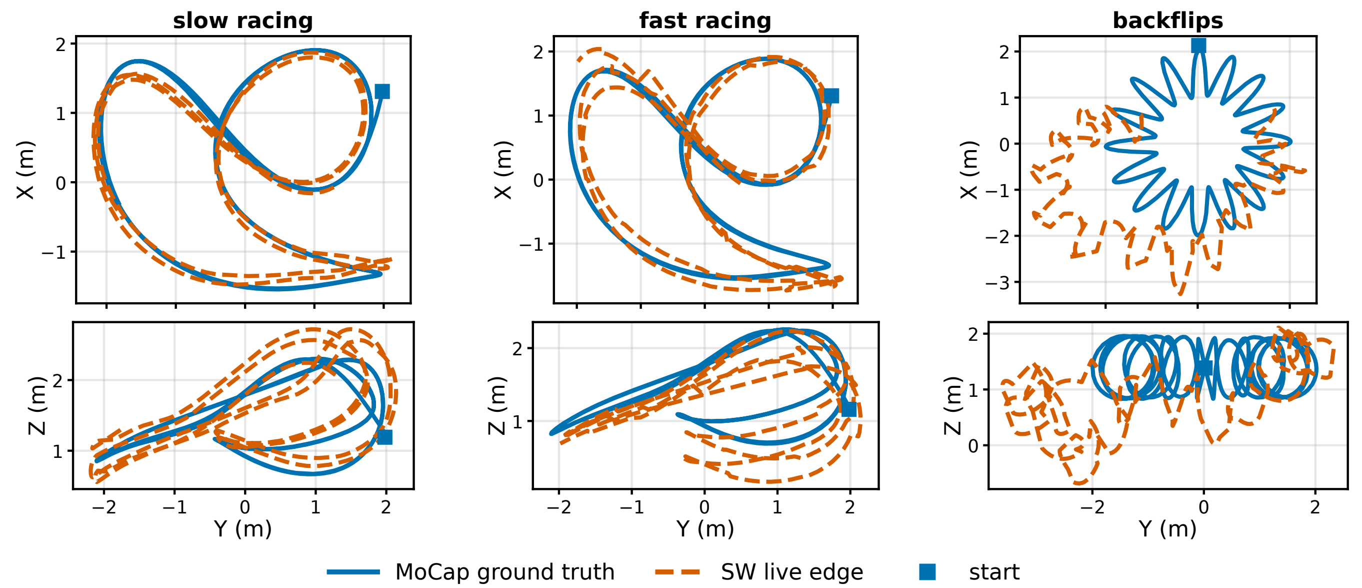

Fig. 3. Trajectory overlays for all three flights (columns): MoCap ground truth vs. the SW live edge, X–Y (top) and Y–Z (bottom) projections sharing the Y axis; start ■. Larger fast-racing deviations reflect higher dynamics ($\omega_{\max} = 9.1\,\text{rad/s}$); on backflips the residual $\sim 6°$ live orientation error attenuates the loop circles while the donut sweep and altitude are followed.

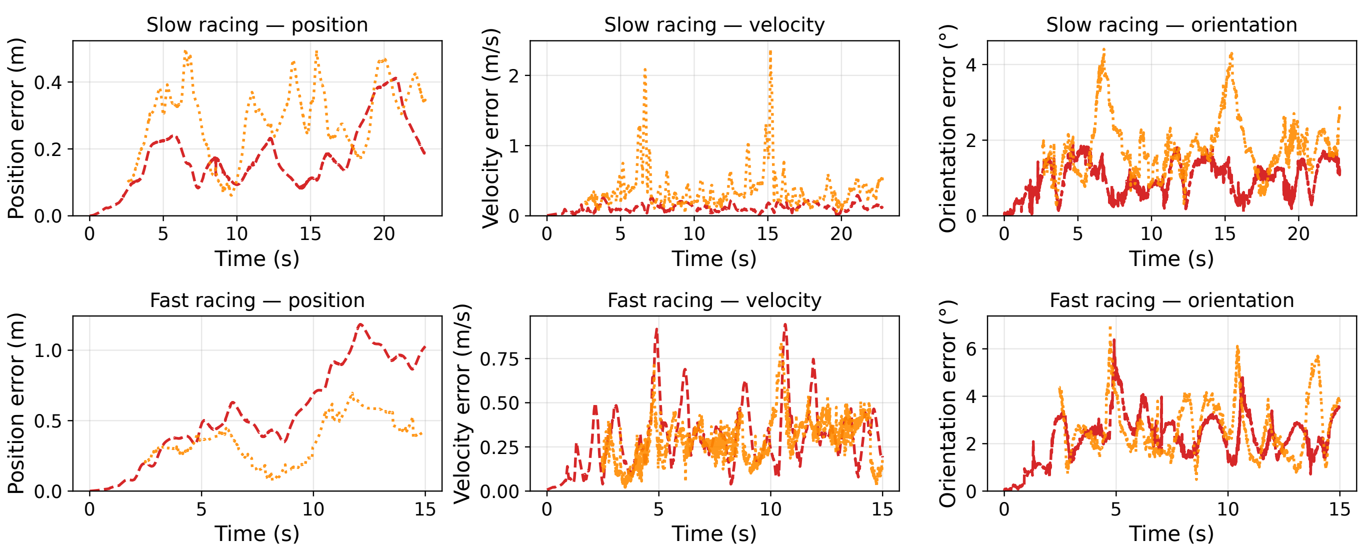

Fig. 4. Position, velocity, and orientation error vs. time for slow racing (top) and fast racing (bottom). Batch estimate (red dashed) and SW live edge (orange dotted). Fast racing position error accumulates monotonically with trajectory length; velocity and orientation errors spike at high-speed, high-dynamics sections.

their toolbox pinned at a fixed commit and every parameter deviation documented in the repository.

Our system on their data. We run our sliding-window solver, with the universal weighting configuration unmodified, on the four public ICINS-2021 indoor flight datasets [20] (TI IWR6843AOP at 10 Hz, Analog Devices ADIS16448 IMU at 409 Hz with onboard barometer, loop-closure-and-scale-corrected VINS pseudo ground truth), using their published extrinsic calibration (locked) and their $\pm 60°$ elevation pre-gate. Their estimators (ekf-rio, unaided yaw; ekf-yrio, Manhattan-world radar yaw aiding) are re-run from their stock configurations on the same bags, and *both* systems are evaluated with the identical causal protocol of Section V (live estimates, start-anchored alignment, same windows and ground truth). Table V reports the headline metrics (a) and their decomposition (b). On position our solver reaches the same order of magnitude as the baselines (0.5–1.9 m, 0.6–1.3% drift). It trails the yaw-aided ekf-yrio on absolute position (best on all four flights), but velocity RMSE 0.07–0.08 m/s matches theirs and the order-of-magnitude vertical gap closes (below). The full 3-DoF orientation column favors our solver (0.51–1.13° vs. up to 10.1°), but this gap is almost entirely *yaw*: our heading is anchored by a pseudo-magnetometer (here the VINS pseudo-GT yaw used as a magnetometer surrogate, in place of the

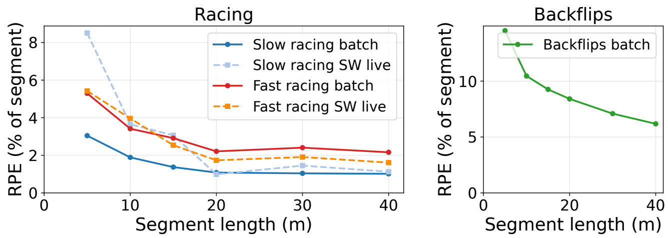

Fig. 5. KITTI-style translational RPE (end-point displacement error as % of segment length) vs. accumulated GT distance. Left: racing bags, batch (solid) and SW live edge (dashed). Right: backflips batch.

MoCap yaw available on our own platform), whereas ekf-yrio uses sensor-only Manhattan-world yaw and ekf-rio is unaided. On the heading-*independent* roll and pitch the three systems are comparable (0.25–$0.74°$, Table V(b)), so the orientation column is not an attitude win; removing our heading aiding confirms this: on flight_1 our unaided yaw drifts to $9.9°$, matching unaided ekf-rio ($10.1°$), while roll/pitch and position are essentially unchanged.

The default RANSAC prefilter (Section IV-B) buys this same-order position, its clearest single demonstration. A single-chip 2-TX array has limited elevation diversity, and a minority of returns carry an elevation-correlated Doppler bias. Doer's reve rejects these with 3D RANSAC (inlier 0.15 m/s); a Huber-robust least squares only down-weights them. Measured against the ground-truth attitude, the per-frame vertical ego-velocity bias is -0.06 to -0.16 m/s under a Huber-only front-end but at most 0.10 m/s in magnitude under RANSAC. With the prefilter *disabled*, our vertical channel accumulates a large systematic error (whole-trajectory absolute trajectory error (ATE) 9.6 m on flight_1, 12–15% vertical drift) that the baselines do not exhibit (theirs stays $\sim$0.2–0.3 m); the RANSAC prefilter removes it at the source on all four flights, cutting whole-trajectory ATE to 0.24–0.76 m (Table V(b)) and bringing position back into the baselines' range. Their vertical accuracy is not altitude aiding either: disabling the barometer in ekf-yrio shifts its vertical drift by $\leq$0.04 m (stock $\sigma_{\text{altimeter}}$=5 m makes it near-negligible), so they too estimate vertical from radar and IMU alone, by the same outlier-rejection mechanism. The single-chip elevation bias is therefore real but not fundamental: an outlier-rejection gap, closed by the same RANSAC stage the baselines use. Under whole-trajectory Umeyama alignment (Table V(b)) all three systems agree to $\leq$1 m, consistent with the baselines' published full-alignment figures (they are faithfully reproduced) and confirming that our default front-end matches them.

Their filters on our data. The reverse port is a *sensor/stack-coupling characterization*, not a head-to-head comparison; it is regime-dependent, and we report the measured outcome rather than a verdict. Here we port ekf-rio and its multi-radar generalization x-rio [22] (run single-radar); ekf-yrio, the yaw-aided single-radar variant Doer benchmarks the ICINS flights with, was the forward-port comparator above. On the slow-racing flight both ekf-rio and x-rio fail: their $\chi^2(3)$ innovation gate rejects 99.7% of the radar ego-velocity updates (301 of 302), so the filter reverts to IMU dead-reckoning and runs essentially unconstrained (>2 km of error for both, almost entirely vertical, $\approx \frac{1}{2}gt^2$). On the faster flight ekf-rio instead survives, degrading to 1.4 m (3.1% drift, $9.0°$): its larger translational excitation produces radar ego-velocities that clear the gate (only 64% rejected), enough to constrain the filter. The amount of radar information admitted thus scales with how strongly motion excites the sparse Doppler, and the common cause is a stack-level mismatch rather than a defect of their method: our best-velocity firmware yields only $\sim$13 points per frame (their RANSAC/σ-gates expect hundreds, and ego-velocity estimation fails outright on roughly half of all scans), there is no hardware radar trigger, and our racing-frame vibration (measured gyro white-noise PSD $2.1°/\text{s}/\sqrt{\text{Hz}}$ in flight) exceeds their process-noise parameter range, making propagation overconfident so the gate then rejects the few updates that survive. This is a design-space split (Section VIII), not a defect of either method: their stack runs near sensor rate on dense clouds, ours spends 0.35–0.70 s per window on sparse Doppler, an accuracy-versus-latency trade.

G. Output Covariance Consistency (NEES)

A factor source for a larger fusion graph must report *honest* covariance. We extract the live-edge joint covariance of $(\mathbf{v}_w, \mathbf{R})$ each window from the converged problem (`ceres::Covariance` over the active support control points, tangent space, marginalization prior included) and compute the normalized estimation error squared (NEES) against MoCap; for a consistent estimator the NEES of each 3-DoF block is χ^2_3-distributed with mean 3. On the racing flights, summarized by the per-window median for *both* channels, the *orientation* NEES is 1.2–2.6 and the velocity NEES 1.4–5.4 (consistent value 3): orientation is essentially calibrated (slightly conservative on slow racing, near-exact on fast, as the weights approximate effective-noise whitening), and velocity is calibrated to mildly overconfident. We report the median, not the mean, because the means (velocity especially) are inflated by a few windows where the finite-difference MoCap velocity reference is unreliable. Both are *calibration checks*, not powered hypothesis tests: consecutive windows are temporally correlated, so the χ^2_3 mean-3 expectation holds only heuristically. On backflips the orientation covariance is overconfident by $\sim$3$\times$ in standard deviation (unmodeled flip-time error) and only 6 of 50 windows survive the ground-truth quality mask (Section V), too few to support a consistency statement, so we report it as a caveat rather than a result. Deployment recipe: inflate the emitted covariance per regime (1$\times$ gentle, 3$\times$ aggressive), or calibrate online from innovations.

H. Ablation Studies

Table VI summarizes key design choices.

Key observations from Table VI. Full-rate IMU (1 kHz) improves position by 2.9$\times$ over 200 Hz, motivating the tight bias priors in `solver_cpp.yaml`. Lever-arm correction helps racing (8–11% position) but has negligible effect on backflips: at Δt_o=0.8 ms the dense spline absorbs the $|\boldsymbol{\omega}\times\mathbf{t}_{bs}| \approx 1$ m/s term through orientation DoF. Schur marginalization reduces fast-racing settled position by 24% (0.412 m $\rightarrow$ 0.312 m); for slow racing the optimal prior scale

TABLE V

CROSS-VALIDATION ON THE PUBLIC ICINS-2021 FLIGHT DATASETS [20], ALL SYSTEMS UNDER OUR CAUSAL PROTOCOL (LIVE ESTIMATE, START-ANCHORED ALIGNMENT). **(A)** HEADLINE METRICS; HEADING AIDING DIFFERS BY DESIGN (SEE TEXT). **(B)** DECOMPOSITION, MAKING THE HONEST STRUCTURE VISIBLE AT A GLANCE: *roll/pitch* IS THE HEADING-INDEPENDENT ATTITUDE RMSE (PER-AXIS RMS, YAW EXCLUDED), SO THE LARGE ORIENTATION GAP IN (A) IS YAW, NOT ATTITUDE; *our drift* SPLITS OUR POSITION ERROR INTO HORIZONTAL AND VERTICAL, BOTH SUB-1.3% UNDER THE DEFAULT RANSAC PREFILTER (SECTION IV-B; THE FOOTNOTE GIVES THE PREFILTER-DISABLED FIGURES FOR CONTRAST); *whole-traj. ATE* RE-EVALUATES EVERY SYSTEM UNDER THE STANDARD EVO/KITTI UMEYAMA ALIGNMENT (NO SCALE) FOR CROSS-CHECKING AGAINST PUBLISHED FIGURES.

(a) Headline: position [m] / velocity [m/s] / orientation [°] / drift [%]

Flight (path)	ekf-rio	ekf-yrio	Ours
1 (142 m)	1.41 / 0.20 / 10.1 / 1.0	**0.39** / **0.08** / 0.70 / **0.3**	0.92 / 0.08 / **0.51** / 0.6
2 (45 m)	0.25 / 0.08 / 1.59 / 0.6	**0.20** / **0.08** / **0.91** / **0.4**	0.51 / 0.08 / 0.94 / 1.1
3 (147 m)	**0.47** / 0.08 / 1.07 / **0.3**	0.48 / 0.08 / 1.09 / **0.3**	1.92 / **0.07** / **0.52** / 1.3
4 (78 m)	0.39 / 0.09 / 3.67 / 0.5	**0.22** / **0.07** / 1.41 / **0.3**	0.92 / 0.07 / **1.13** / 1.2

(b) Decomposition

	Roll/pitch RMSE [°]			Our drift [% path]		Whole-traj. ATE [m]		
Flight	ekf-rio	ekf-yrio	Ours	horiz.	vert.	ekf-rio	ekf-yrio	Ours
1	0.24	0.25	0.25	0.16	0.63	0.87	0.23	0.46
2	0.57	0.59	0.55	0.14	1.13	0.13	0.10	0.24
3	0.25	0.26	0.27	0.30	1.28	0.36	0.26	0.76
4	0.74	0.74	0.74	0.26	1.15	0.30	0.15	0.46

Prefilter *disabled* (Huber-only): our vertical drift 12–15%; whole-traj. ATE $9.57/2.88/10.86/5.48$ m (flights 1–4).

is 10^{-7} (near-zero), so both variants give equivalent results; the benefit is captured in Table VII instead. Window duration has little effect on either racing regime: slow stays at live 0.30–0.34 m across 1.5–5 s, and fast gives essentially equal orientation at 2 s and 3 s (settled 2.31° vs 2.35°) with 3 s modestly better on position (live drift 0.98 vs 0.88%). We keep 3 s as the default for that small position gain, trading latency. We lock pitch at its measured value for all sliding-window runs; even in the batch ablation, locking is marginally better than freeing it (0.157 vs. 0.169 m on slow racing), consistent with the pitch being only weakly observable (Section VII-C).

VII. LIMITATIONS AND NEGATIVE RESULTS

A. Statistical Scope

Each flight regime is evaluated on a *single* representative recording, so the headline numbers carry no run-to-run error bars. The held-out flights (Section VI-E), the cross-platform validation (Section VI-F), and the deterministic, bit-identical re-runs broaden the evidence, but they do not substitute for repeated trials: re-runs add reproducibility, not statistical power. Reporting mean $\pm$ std over several recordings per regime is the natural strengthening, and is left to future data collection.

B. Heading Sensor Requirement

Yaw is a gauge freedom for Doppler: rotating the entire trajectory about the world gravity axis leaves all radial velocity measurements unchanged, so absolute heading cannot be recovered from radar and IMU alone. A magnetometer resolves this and is standard on virtually all flight platforms; here MoCap yaw serves that role. Because that MoCap yaw is noise- and bias-free (unlike a real magnetometer), the heading-aided orientation numbers are correspondingly optimistic; the heading-*independent* roll/pitch decomposition (Table V(b)) and the unaided-yaw check (Section VI-F) isolate attitude accuracy from this idealization. This is a sensor-architecture requirement, not an open research problem.

C. Extreme Dynamics: Backflips

Batch. The batch solver handles backflips only with Δt_o=0.8 ms (10$\times$ denser than racing, 22,630 knots over 18 s) and locked extrinsics (free pitch drifts to $+18°$ due to dense under-constrained orientation DoF). At this density the batch orientation is *bistable* (detailed below): under small timestamp perturbations it oscillates between $\sim$8° and $\sim$25° regimes, so we do not report a single batch figure and take the sliding-window result as the trustworthy one.

Sliding window. The fixed-lag smoother initially appeared fundamentally unsuited to backflips: with the inconsistent prior and a (silently inactive) position anchor, the best result was 2.53 m/10.8° settled. A chain of three fixes revised this verdict. (i) The consistent unscaled prior (Section IV-E) restores inter-window information transfer. (ii) A soft position anchor to the dead-reckoned initialization ($\lambda_{\text{pos-init}}$=0.5 per CP) bounds the position random walk that sparse, rate-inflated radar cannot prevent within a 3 s window (without it position diverges to 8 m; on racing bags the same anchor is *harmful*: a flip-regime rescue, not a general ingredient). (iii) The dynamics-adaptive weighting law (Section IV-B), in particular rate-adaptive gyroscope trust, which on this bag alone is worth $\sim$3° of orientation at zero position cost, plus the flip-regime Doppler correction b=-1.5 m/s. Together these reach **1.67 m / 5.0° settled and 1.55 m / 6.4° live at $\sim$0.6 s per window** (Table III; Fig. 3). The default RANSAC prefilter (Section IV-B) is neutral here (backflip frames are already sparse and mostly fall below its five-return floor, so they bypass it), and the small change from the Huber front-end ($1.51 \rightarrow 1.55$ m live) is within run-to-run noise.

TABLE VI
SELECTED ABLATION RESULTS (BATCH SOLVER, SLOW_RACING UNLESS NOTED)

Configuration	Pos (m)	Ori (°)
IMU rate (slow_racing batch, no lever arm†)		
200 Hz	0.525	1.64
1000 Hz (default)	**0.183**	**1.02**
Lever-arm correction $\boldsymbol{\omega}\times\mathbf{t}_{bs}$ (batch)		
No correction — slow racing	0.183	1.02
— fast racing	0.862	2.73
— backflips	1.814	8.64
With correction — slow	**0.169**	**1.02**
— fast	**0.768**	**2.64**
— backflips	**1.814**	**8.44**
IMU preintegration vs. raw factors (slow_racing batch)		
Forster TRO-2017 preint. at 100 Hz	0.288	6.40
Raw gyro/accel at 1 kHz (default)	**0.169**	**1.02**
Orientation regularization (backflips batch)		
Min-ω (λ_ω=0.001)	1.816	8.62
Min-α, λ_α=0.1 (default)	**1.814**	**8.44**
Min-α, λ_α=1.0	2.069	83.8
SW window dur. (slow_racing SW, settled / live edge)		
1.5 s	0.292 / 0.299	1.51 / 2.01
3.0 s (default)	**0.293 / 0.310**	**1.53 / 1.97**
5.0 s	0.313 / 0.341	1.49 / 2.03
SW window dur. (fast_racing SW, settled / live edge)		
2.0 s	0.380 / 0.443	2.31 / 2.95
3.0 s (default)	**0.312 / 0.399**	**2.35 / 2.88**
Marginalization (fast_racing SW, settled)‡		
None (boundary prior only)	0.412	2.71
Schur complement	**0.312**	**2.35**
Extrinsic optimization (slow_racing batch)§		
Pitch δ optimized (default)	**0.169**	**1.02**
Pitch δ locked	0.157	1.02

Batch rows (IMU-rate, lever-arm, ori-reg, extrinsic) isolate one design choice at a fixed pre-RANSAC operating point; the RANSAC default shifts batch absolutes (slow batch 0.169→0.198 m, fast 0.768→0.64 m, the Section VI batch figures) without changing orderings. The SW rows (window duration, marginalization) are under the RANSAC default. All runs use the sensor-only init (Section IV-D); the bias seed shifts batch absolutes comparably, again without changing orderings. †Lever arm disabled for isolation; see Lever-arm rows for the current default (0.169 m/1.02°). ‡Marginalization shown for `fast_racing`; on slow_racing the near-zero optimal prior makes Schur and boundary-only nearly equivalent. §C++ solver supports only 1-DoF pitch offset (roll/yaw unobservable from Doppler; free 3-angle optimization tested in Python solver showed 6–8° orientation runaway; see Section VII-E).

Why Δt_o=8 ms in the window. At the batch density Δt_o=0.8 ms a 3 s window has ∼11,250 orientation DoF against ∼9,000 gyroscope constraints (1.26 DoF/constraint); the per-window Jacobian is rank-deficient and the solver freezes near its initialization. At 8 ms the window is well-conditioned (62.5 Hz model bandwidth comfortably covers the 10–20 Hz flip dynamics). The same density is what makes the *batch* problem bistable (the ∼8°/∼25° oscillation noted above): the final cost *anti-correlates* with accuracy (the underconstrained problem overfits the wrong global orientation), so single-run comparisons at this density are not trustworthy.

Attributing the z-axis failure, and what b really is. A visual trajectory check (numeric RMSE conceals it) showed the SW estimate sinking in altitude. Disabling the position anchor exposes the raw behaviour: a near-constant -0.57 m/s world-z drift (-8 m over the flight). Sweeping the per-point correction $v_{\mathrm{corr}} = v - b\,u_{s,z}$ produces a linear family of z-error curves and large negative b flattens the drift, so we initially interpreted b as the antenna-fixed elevation bias of the 2-TX array. A joint calibration experiment *disproves* this interpretation: (i) with extrinsic pitch *locked* at its measured value, level-flight racing degrades catastrophically under any explicit b (fast-racing live position $0.39 \to 3.2 \to 6.6$ m for b=0, -0.5, -1.0): a true antenna-fixed bias would be tolerated, since nothing else can absorb it; and (ii) on backflips the optimum keeps growing monotonically past the WLS-measured static bias of -0.5 m/s (through -1.5 to -2.0), into physically implausible territory. We conclude b is a *flight-regime error proxy*: it absorbs an attitude-correlated Doppler systematic during flips (most likely intra-frame motion distortion) that happens to project like an elevation term, and we cap it at b=-1.5 rather than chase the monotone trend without a mechanism. Relatedly, extrinsic pitch is a *plateau*: over a 3×3 grid over (locked pitch, b) the RMSE is indistinguishable across 26.5–28.5° on backflips, and locking at 27.5° versus online sliding-window self-calibration is RMSE-neutral on racing. We freeze it rather than estimate it online because free pitch is only weakly observable and *init-dependent in both* the windowed solve and the full-trajectory batch: the converged value tracks the initialization (a 25° seed gives 27°, a 30° seed gives 36° on both racing bags, diverging from a saddle near 23°). The earlier "self-calibrated +2° absorbs the elevation bias" reading is retired; we instead fix pitch at the physically measured as-built angle (27.5°, inclinometer; Section VI), a few degrees below the 30° CAD design (the 3D-printed, loosely-fastened fixture accounts for the difference).

Remaining gap. The residual ∼6° live orientation error during flips attenuates the reconstructed loop circles relative to ground truth. This is not a spline-bandwidth limit (Section VII-E: denser knots do not help); the residual lever is the radar measurement model, whose intra-frame Doppler smearing during flips is best addressed by *per-chirp timestamps* rather than knot density, which we leave to future work.

D. Prior Scale Sensitivity as a Symptom of Inconsistent Marginalization

Our initial marginalization implementation selected all residuals touching marginalized *or* boundary variables, which, through the shared bias block, baked every IMU factor of the window into the Schur complement, conditioned on interior states, and double-counted factors re-added in the next window (Section IV-E). The resulting overconfident prior had to be deflated by a hand-tuned scale, and the required scale varied by orders of magnitude across flight regimes with a harmful intermediate region. We document this behaviour because the symptom pattern (no universal prior weight, non-monotone sensitivity, robust losses on the prior acting backwards) is a useful *diagnostic for marginalization inconsistency* in sliding-window estimators generally. Table VII shows the sweep measured with the inconsistent prior; Fig. 6 plots the orientation sweep.

Slow racing is best served by a near-zero prior (10^{-7}): with gentle dynamics, each 3 s window is self-sufficient and

TABLE VII
PRIOR SCALE SENSITIVITY (SW, LIVE-EDGE RMSE)

`scale`	Slow racing Pos (m)	Slow racing Ori (°)	Fast racing Pos (m)	Fast racing Ori (°)
2×10^{-4}	0.543	2.16	**0.825**	**3.64**
10^{-5}	—	—	0.942	3.58
10^{-6}	0.442	2.21*	0.951	3.57
10^{-7} (slow default)	**0.338**	**2.08**	1.279	3.57
10^{-8}	0.336	2.08	1.344	3.57

*Orientation local maximum: 2.21° > 2.16° (2e-4) and 2.08° (1e-7). —: scale not evaluated for this bag.

Measured under the inconsistent-prior diagnostic configuration (predates the RANSAC default); the sensitivity *pattern*, not the absolutes, is the result.

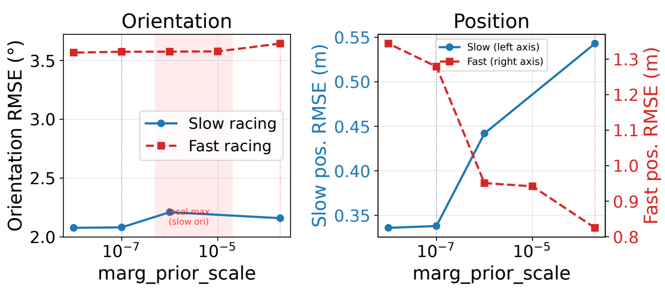

Fig. 6. Live-edge orientation (left) and position (right, dual y-axes) RMSE vs. `marg_prior_scale`. Orientation (left): slow racing shows a local maximum at 10^{-6}; fast racing orientation is insensitive to scale. Position (right): slow racing worsens monotonically with increasing scale while fast racing improves sharply: the two bags require opposite scale regimes.

additional inter-window coupling is harmful. Fast racing instead needs the prior (2×10^{-4}) for inter-window position continuity; without it (scale= 10^{-7}) fast-racing live position degrades from 0.825 m to 1.279 m. The two bags thus want opposite scales, and the non-monotone slow-racing orientation maximum at 10^{-6} (Table VII) shows no single scale is universally optimal.

The information asymmetry in the inconsistent S (orientation eigenvalues $\sim5.5\times10^{14}$, position $\sim10^{4}$) meant no single scalar could address both modalities; eigenvalue clipping and Cauchy robustification of the prior also failed (the latter counter-productively down-weights the bag that needs the prior most, because aggressive dynamics produce larger prior residuals).

Resolution. With the Markov-blanket factor-selection rule of Section IV-E, the prior is a true marginal: a single *unscaled* prior (scale = 1) is optimal or near-optimal on all three bags simultaneously, the prior's Mahalanobis residual at each solution is $\mathcal{O}(1)$, and the entire per-mission tuning table becomes obsolete. The fix also improves accuracy: fast-racing settled position improves from 0.727 m (tuned inconsistent prior) to 0.701 m (unscaled consistent prior). Both figures isolate the prior-consistency effect at a fixed pre-RANSAC operating point; the RANSAC default brings the same metric to 0.312 m (Table III). Stride-boundary position jumps in the settled trajectory also disappear (settled velocity RMSE 0.89 → 0.21 m/s on slow racing).

E. Other Negative Results

Adaptive knot density. We tested whether denser or non-uniform orientation knots reduce the residual backflip attitude error; they do not. At Δt_o=8 ms the orientation spline already represents the MoCap attitude to 1.28° RMSE *during* the flips, and its angular-velocity tracking of the gyro stream is 0.295 rad/s RMS, flat across $\Delta t_o \in \{16, 8, 4\}$ ms and sitting at the rate-independent vibration floor (0.17 rad/s). The flip orientation gap is therefore *not* a spline-bandwidth limit: the spline can represent the motion, but the optimizer does not rotate to it (the gyro residual correlates with $|\boldsymbol{\omega}_b|$ at $+0.94$ against the *solved* trajectory, and open-loop gyro dead-reckoning is 7.8°, better than the 9.2° this fusion reaches *at the diagnostic configuration*, before the rate-adaptive gyro trust of Section VII-C brings it to the 6.4° headline of Table III). The lever is measurement weighting and robustness (Section VII-C), not knot density: the dynamics-adaptive gyro trust closes most of the gap, *disproving* the adaptive-knot-density hypothesis for this regime and obviating the non-uniform-basis extension.

IMU preintegration. Replacing per-sample gyro/accel factors with Forster TRO-2017 [23] preintegrated factors at 100 Hz reduces constraint density from 8:1 (samples per knot) to 1:1, degrading orientation RMSE from 1.02° to 6.40° and position RMSE from 0.169 m to 0.288 m on slow racing. The smaller Jacobian iterates faster, but the accuracy loss is unacceptable.

GNC (Graduated Non-Convexity). Geman-McClure loss annealing [24] on radar residuals improved results only when extrinsics were incorrect (masking calibration errors as outliers); with correct calibration it degraded both bags and added 3× runtime.

Full extrinsic optimization. The C++ solver exposes only the 1-DoF pitch offset δ_{pitch} (Section IV-A): Doppler cannot observe roll or yaw, and an earlier Python solver that freed all three angles let roll and yaw drift 5–7° into the unobservable modes as a residual trash can, corrupting orientation by 6–8°. The C++ design avoids this by construction.

VIII. CONCLUSION

Can a single-chip mmWave radar plus IMU provide usable 6-DoF state estimation? Yes, as a *velocity and orientation* source: at the live leading edge, 0.48 m s^{-1}/2.0° on gentle racing, 0.32 m s^{-1}/2.9° on aggressive racing (0.35–0.70 s per window on a laptop), confirmed on held-out flights and with an approximately calibrated (NEES) orientation covariance on racing. Absolute position is not observable from Doppler+IMU and drifts with distance (0.6–0.9% on racing, 2.9% under tumbling); yaw requires a heading sensor. Within those structural limits, *good results took three things*: (1) a *consistent* fixed-lag smoother, with a Markov-blanket factor-selection rule that makes the marginalization a true marginal, eliminating regime-dependent prior tuning whose symptom pattern we document as a reusable diagnostic; (2) a *dynamics-adaptive trust allocation* across the measurement streams (rate-adaptive gyroscope weight, rate-inflated radar/accelerometer noise, per-point SNR weighting), one configuration from hover to 10 rad s^{-1}, with the velocity–orientation trade made explicit as a measured Pareto curve; and (3) *characterized sensor systematics*, including the finding that the apparent elevation bias under tumbling is a flight-regime error proxy, not an antenna constant, and that the extrinsic pitch is weakly observable, best fixed at its measured value rather than estimated online. Notably, the largest accuracy gains came from diagnosis-led re-weighting of existing measurements, not from added model capacity: a representation-level study showed the orientation spline already tracks $10\,\mathrm{rad\,s^{-1}}$ flips at the gyroscope noise floor, disproving our own adaptive-knot-density hypothesis before implementation.

Open limitations: (1) loop-scale position geometry through flips is bounded by during-flip radar data quality (Doppler noise $15\times$ the level-flight core), demonstrated by an asymmetric factor-split ablation; (2) ground truth itself degrades during flips (MoCap occlusion), bounding what can be measured there; and (3) cross-validation against the EKF-RIO family (Section VI-F) reveals a design-space split. With our default RANSAC front-end our solver reaches the same order of magnitude on position on their public flights (0.5–$1.9\,\mathrm{m}$, trailing the yaw-aided baseline), matches their roll/pitch, and leads on heading-aided orientation; the prefilter closes the single-chip elevation-bias gap. In reverse, their stock filters, starved by our sparse $\sim$13-point clouds and racing-frame vibration, reject up to 99.7% of radar updates and dead-reckon on slow racing (degrading, not diverging, on the faster flight). Each stack is engineered for its own regime.

Future directions follow from these results. An incremental, iSAM2-style back-end would turn the demonstrated real-time feasibility (Section VI-B) into the onboard margins a deployed factor source needs. Per-chirp radar timestamps target the residual during-flip orientation error, which our negative results place in the measurement model rather than the spline (Section VII-C). Repeated recordings per regime would replace single-flight evaluation with powered statistics (Section VII-A), and the heading weight could be auto-adapted from a yaw-observability proxy rather than set per regime (Section IV-B).

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